Problem Set: Nonlinear Programming

Instructions

In Problems 1–3, you will implement three numerical methods for solving nonlinear equations:

- Bisection Method
- Fixed Point Iteration
- Newton's Method

For each of Problems 1–3:

- Write your own implementation from scratch (no use of built-in solvers like `uniroot()` or `nleqslv()`).
- Use a convergence criterion of $|x_{k+1} - x_k| < 10^{-6}$.
- Include a plot of the function over an appropriate interval.
- Report the final root estimate, number of iterations, and whether convergence was successful.

For Problems 4–5, solve analytically using the KKT conditions and show your work.

Problem 1: Bisection Method

Let

$$f(x) = \cos(x) - x$$

Use the bisection method to solve $f(x) = 0$ on the interval $[0, 1]$.

Tasks

a. Confirm that $f(0) \cdot f(1) < 0$.
b. Implement the bisection method.
c. Plot $f(x)$ over $[0, 1]$ and mark the root.
d. Report: number of iterations, final estimate, and value of $f(x)$ at the root.

Comment: Explain why convergence is guaranteed given the function.

Problem 2: Newton's Method

Let

$$f(x) = x^3 - 2x + 2$$

Tasks

a. Implement Newton's method.
b. Use $x_0 = 0$ and $x_0 = -1.5$ as two different initial values.

c. Plot $f(x)$ and its derivative over $[-3, 3]$.
d. Report the outcome of the iteration for each starting point.
e. Compare the convergence (or divergence) behavior. What role does the choice of initial guess play?
f. Why does the method fail to move when starting at $x_0 = 0$? explain.

Problem 3: Pollution Abatement under an Emissions Cap

A firm chooses how much output q to produce and how much pollution abatement a to undertake to maximize profit. It sells output at a price of 100 per unit and incurs production costs of $C(q) = 2q^2$. The firm can reduce pollution by using cleaner technology or filters, at an abatement cost of $A(a) = \frac{1}{2}a^2$.

Each unit of output generates one unit of pollution before abatement, and each unit of abatement reduces emissions by one unit. A regulator limits net emissions to a cap of E. The firm must choose nonnegative output and abatement, $q \geq 0$ and $a \geq 0$. Consider how the firm adjusts these choices to comply with a cap of $E = 20$ and how a tighter cap affects its profit.

Questions:

a. Lagrangian: Construct the Lagrangian for the problem, introducing a multiplier λ on the emissions constraint, and μ_q and μ_a on the non-negativity constraints.

b. Derive the KKT conditions (stationarity, complementary slackness, primal/dual feasibility).

c. Solution:

- Solve for the unconstrained optimum (q^u, a^u) and show whether the emissions constraint binds.
- Solve for the constrained optimum (q^*, a^*) when the cap is $E = 20$.

d. Verify that the second-order conditions for a maximum are satisfied at the constrained optimum.

e. Interpretation of λ:

- What is the economic meaning of λ^* in this problem?
- How does it relate to the marginal cost of tightening the emissions cap?

f. Comparative statics: Suppose E decreases by one unit (stricter regulation). Use λ^* to predict the change in the firm's profit.

g. For a required output level $\bar{q}$, formulate the firm's cost-minimization problem subject to the production requirement and emissions cap. Derive its cost function, then show how the firm chooses output to maximize profit.

Problem 4: Product Mix

A firm chooses quantities x and y of two products to maximize profit:

$$\max_{x,y} \ \pi(x, y) = 12x - x^2 + 4y - y^2$$

Each unit of either product uses one unit of production capacity. The firm has three units of capacity, and quantities must be nonnegative:

$$x + y \leq 3, \qquad x \geq 0, \qquad y \geq 0.$$

Tasks

a. Find the profit-maximizing quantities when the capacity constraint is absent. Are these quantities feasible with the capacity constraint?

b. Write the Lagrangian using multipliers λ, μ_x, and μ_y on $3 - x - y \geq 0$, $x \geq 0$, and $y \geq 0$, respectively. Derive all KKT conditions: stationarity, primal feasibility, dual feasibility, and complementary slackness.

c. Use the KKT conditions to solve the problem. If a case is impossible, state the specific condition it violates.

d. Report the optimal quantities, profit, and all three multipliers. Verify every KKT condition at your solution. Use the Hessian of the profit function and the shape of the feasible set to explain why this is the unique global maximum.